Semantic Mapping

NICAR 2026

Columbia Journalism School | Tow Center for Digital Journalism

Open in GitHub Codespaces

Clustering and Mapping Text Data

Set up a codespace:
tinyurl.com/nicar-embed

API KEY
tinyurl.com/TOW-NICAR

Today's Hosts

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Columbia Journalism School

TOW CENTER FOR DIGITAL JOURNALISM

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Columbia Journalism Review.

C.J. Robinson is a reporting and research fellow at the Tow Center for Digital Journalism. He graduated from Columbia Journalism School with an MS in data journalism.

Grok Is Now Editing Itself

FEBRUARY 12, 2026 *By* C.J. ROBINSON

More and more, Elon Musk's chatbot is both suggesting and approving changes to Grokipedia, his Wikipedia clone.

Too Many FOIA Requests, Too Little Transparency

FEBRUARY 5, 2026 *By* C.J. ROBINSON

FOIA's systemic problems meet partisan robo-requests.

When the Pink-Slime Researchers Become the Pink Slime

NOVEMBER 20, 2025 *By* DHRUMIL MEHTA AND C.J. ROBINSON

X Is Using AI Fact-Checkers

NOVEMBER 6, 2025 *By* C.J. ROBINSON

Eight AI bots contribute about 10 percent of the platform's "Community Notes."

Why Economists and Doctors Are Monitoring Local News

OCTOBER 28, 2025 *By* C.J. ROBINSON

A reflection on the New(s) Knowledge Symposium.

Cuomo Deepfaked Himself. You Can Too.

OCTOBER 9, 2025 *By* C.J. ROBINSON

Aisvarya Chandrasekar

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Columbia Journalism Review.

Aisvarya Chandrasekar is a journalist with the Tow Center.

AI Chatbots Can Search the Web—So Why Don't They?

JANUARY 27, 2026 *By* AISVARYA CHANDRASEKAR AND KLAUDIA JAŹWIŃSKA

Why AI platforms should be accountable for misrepresenting breaking news

How AI Browsers Sneak Past Blockers and Paywalls

OCTOBER 30, 2025 *By* AISVARYA CHANDRASEKAR AND KLAUDIA JAŹWIŃSKA

OpenAI's Atlas and Perplexity's Comet present a new challenge for media companies that want to limit AI access.

Journalists Need Their Own Benchmark Tests for AI Tools

SEPTEMBER 18, 2025 *By* KLAUDIA JAŹWIŃSKA AND AISVARYA CHANDRASEKAR

The performance tests used by AI companies don't measure what matters in the newsroom.

Why AI Models Are Bad at Verifying Photos

AUGUST 26, 2025 *By* AISVARYA CHANDRASEKAR AND KLAUDIA JAŹWIŃSKA

"You don't know when it's just making stuff up."

'Automating the Boring Stuff' with Pollfinder.ai

JUNE 19, 2025 *By* AISVARYA CHANDRASEKAR

Building AI-powered tools to help newsroom researchers collect polling data.

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Today's Plan

🎤 Semantic Maps

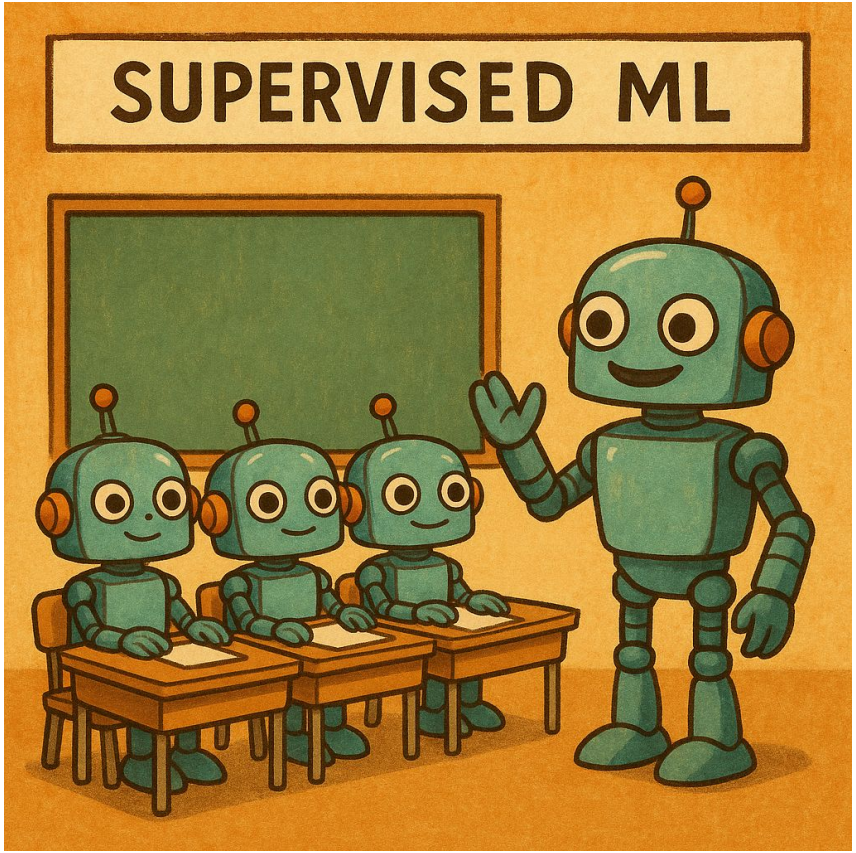
- Making text into vectors
- Dimensionality Reduction
- Examples

💻 Explore: Trump Tweets

💻 Make Your Own

Today's Objectives

- ✓ Understand what **text embeddings** from LLMs are and how they are **visualized.**
  - Know that different methodological choices can result in different maps (which LLM you use, choices in dimensionality reduction
- ✓ Discover what kinds of insights can be produced from **semantic maps** and where they might be helpful in your journalism.
- ✓ Run the code to **generate your own** semantic maps**.**


SUPERVISED ML

UNSUPERVISED ML

Text as Vectors

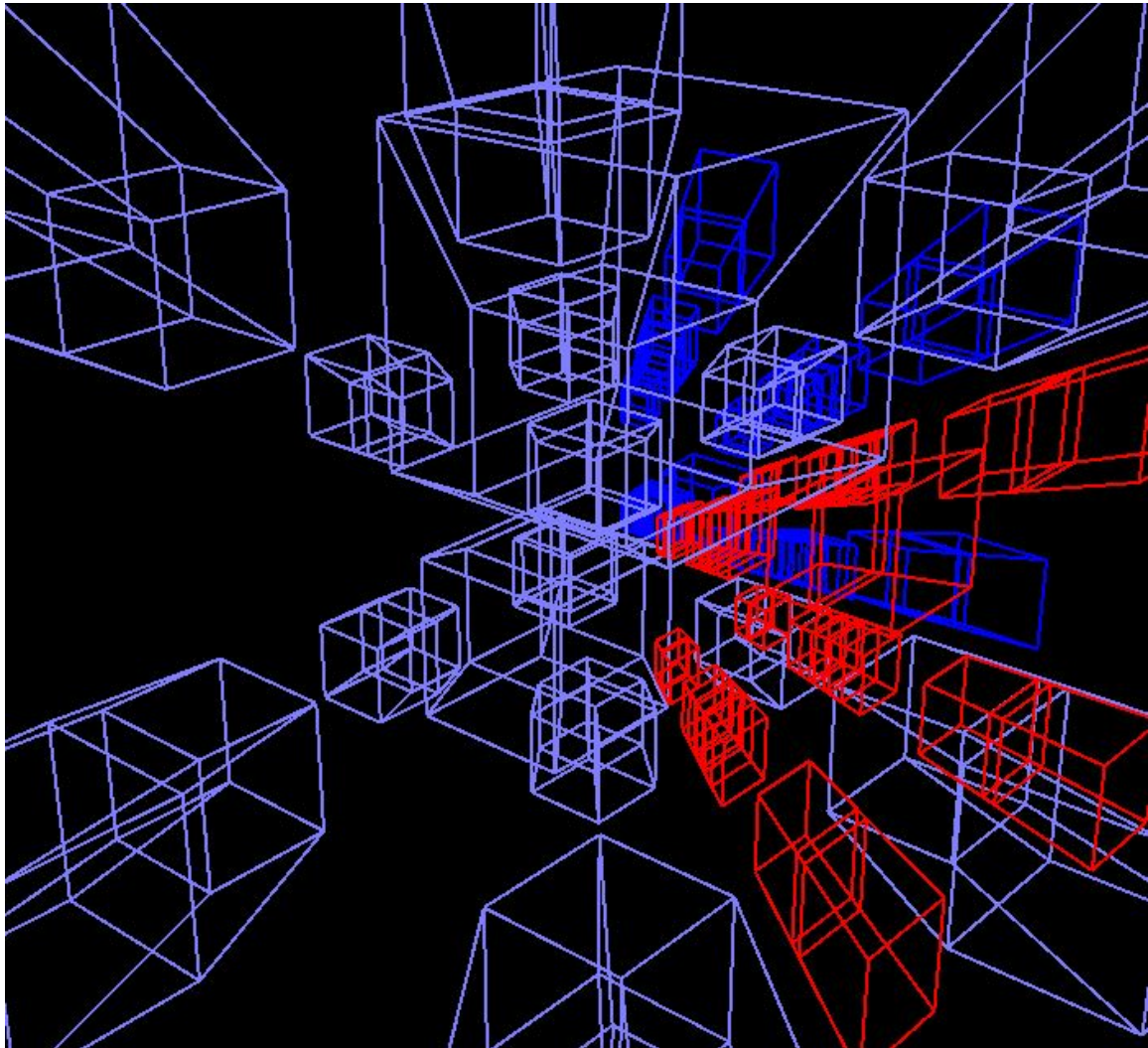
DOCUMENTS ARE MODELED AS **VECTORS**

in

✨✨

MULTIDIMENSIONAL SPACE

✨✨



Text as Vectors (old way)

THESE **VECTORS** in ✨✨ **MULTIDIMENSIONAL SPACE** ✨✨ARE A

BAG OF WORDS model

Because order is not preserved.

BAG OF WORDS

A bag of words is a technique in natural language processing that converts text into numerical format by creating a vocabulary of unique words and counting their occurrences. Ignoring word order, each word frequency serves as a feature in a feature vector.

Raw Text

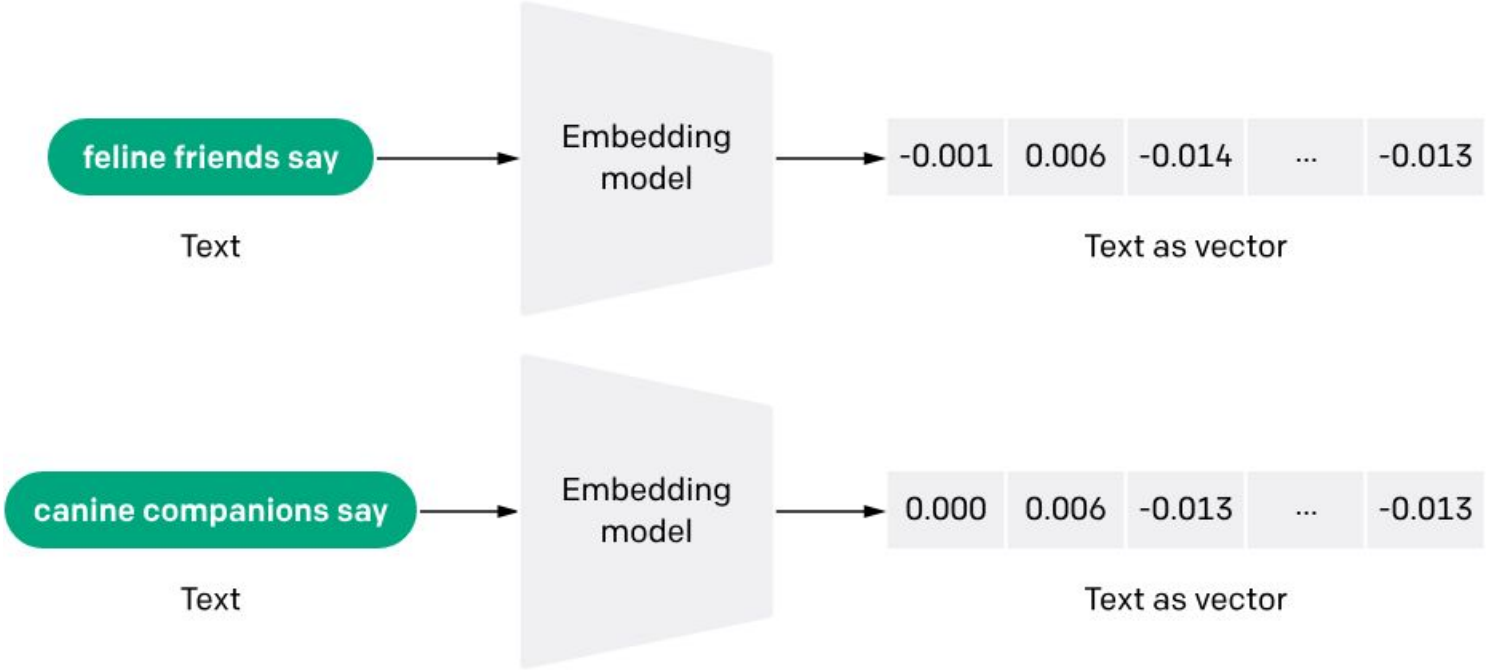
1. "The cat is on the mat."
2. "The dog sat on the mat."

Bag Of Words

	the	cat	is	on	mat	dog	sat
Sentence	2	1	1	1	1	0	0
Sentence	2	0	0	1	1	1	1

aiflashcards.com

Text as Vectors (new way)

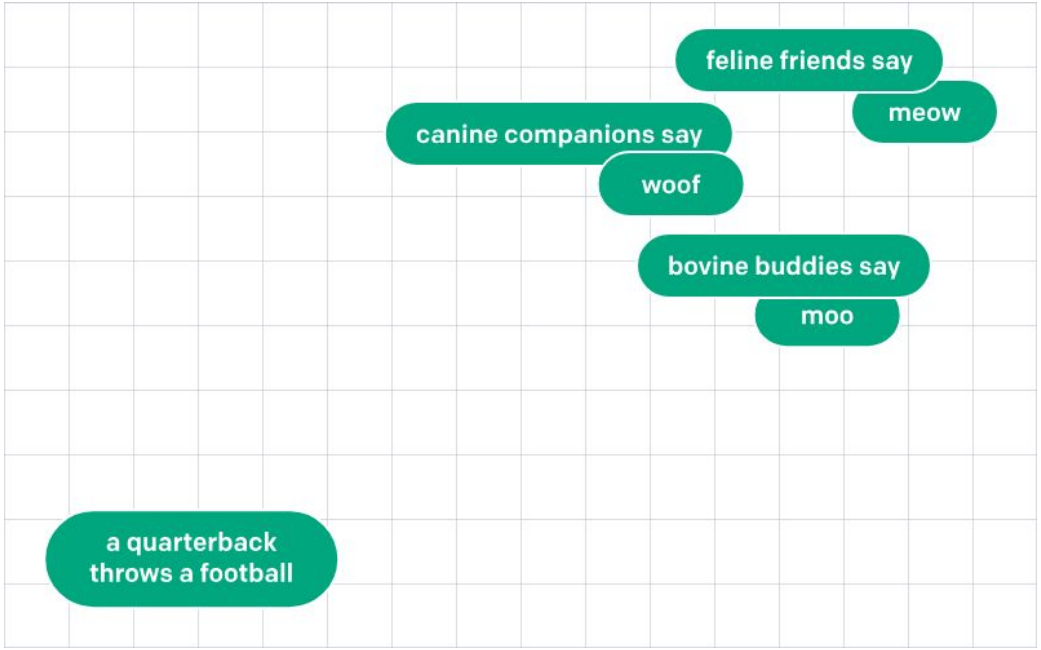


https://openai.com/blog/introducing-text-and-code-embeddings

Text as Vectors (new way)

THESE **VECTORS** in ✨✨ **MULTIDIMENSIONAL SPACE** ✨✨ARE A

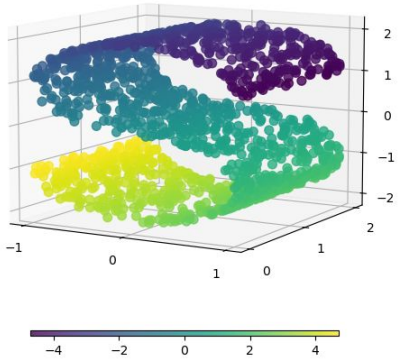
Embeddings from a **LARGE LANGUAGE MODEL**



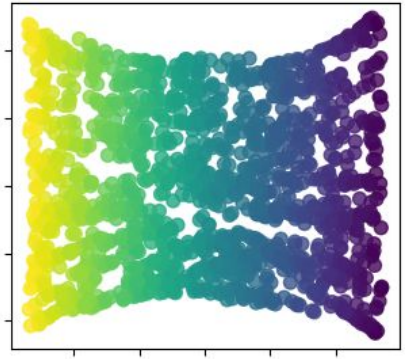
https://openai.com/blog/introducing-text-and-code-embeddings

Dimensionality Reduction

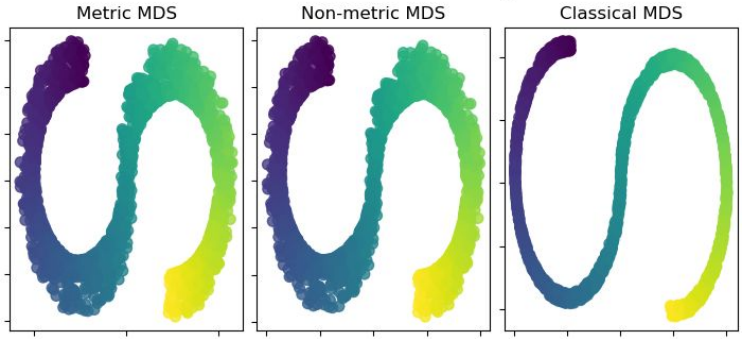
Original S-curve samples



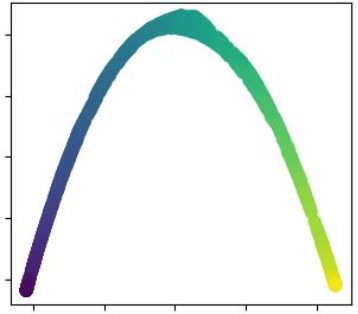
Isomap Embedding



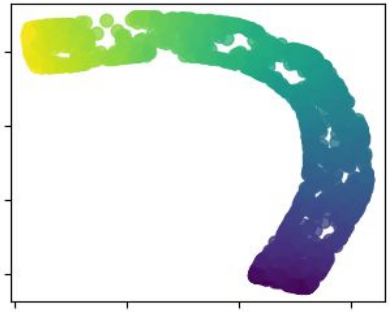
Multidimensional scaling



Spectral Embedding



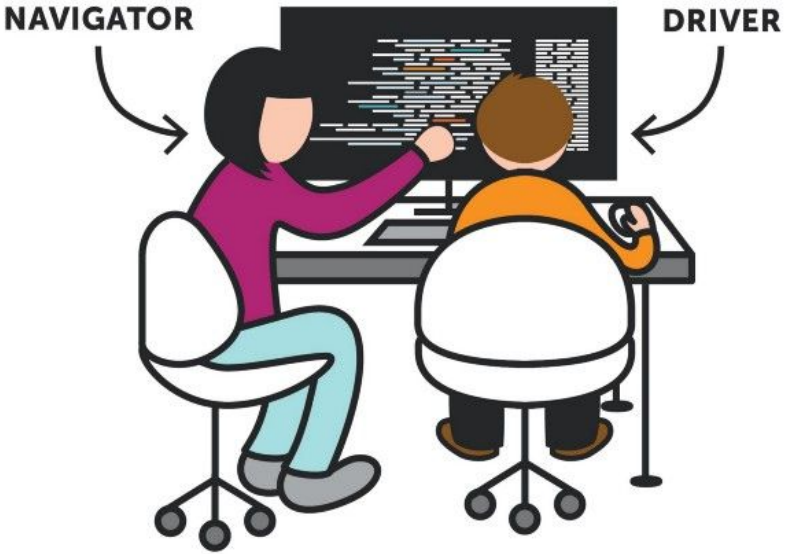
T-distributed Stochastic Neighbor Embedding



Examples

- **Pink Slime (Metric Media) | Tow Center for Digital Journalism**
  - https://dhrumilmehta.com/pink-slime-semantic-map/?filename=wisconsin.csv&search=&start=Tue+2024-10-08&end=Mon+2024-11-11
- **FOIA Scrolly (not yet published)**
- Trouble in Wyoming - Columbia Journalism Review
  - https://dhrumilmehta.com/cowboy-state-daily/
- Courier Newsroom Spent Big on a Meta Ad Blitz in October - Columbia Journalism Review
  - https://dhrumilmehta.com/pink-slime-semantic-map/?filename=courier_ads.csv&search=dollars&start=Wed+2023-03-29&end=Sun+2023-07-23
- **OpenAI's GPT Is a Recruiter's Dream Tool. Tests Show There's Racial Bias - Bloomberg**
- Are you in TikTok's cat niche? What 121,000 videos reveal. - Washington Post

PAIR PROGRAMMING



💻 Explore:
Trump Tweets:
tinyurl.com/trump-tweet-map

Discussion

Make Your Own

API KEY:
tinyurl.com/TOW-NICAR

Git Repo:
tinyurl.com/nicar-embed

grokipedia.com

Grok Edits People

TOTAL ARTICLES
6,092,140

TOTAL APPROVED EDITS
837,338

Grok has approved an edit for **Dave Smith**

Go to edited article

ORIGINAL TEXT

He hosts the podcast ''Part of the Problem'', launched to discuss politics and current events from a libertarian perspective, often incorporating philosophical and comedic elements. Smith also co-hosts the comedy podcast ''Legion of Skanks''. He ha…

See more

SUGGESTED EDIT

This section could include notable people people who have been on his podcast, Part of The Problem, including Tucker Carlson, Candace Owens, Vivek Ramaswamy, RFK jr, Nick Fuentes, Darryl Cooper, Scott Horton, and Michael Malice just to name a few.…

See more

✓ GROK FEEDBACK

Web searches confirmed that the suggested guests—Tucker Carlson, Candace Owens, Vivek Ramaswamy, RFK Jr., Nick Fuentes, and others like Michael Malice—have appeared on "Part of the Problem," with direct evidence from YouTube episodes, podcast platforms like Spotify and Apple Podcasts, and transcripts on sites like singjupost.com. For example, specific episodes feature Tucker Carlson (e.g., episode 1323), Candace Owens (episode 1325), Vivek Ramaswamy (multiple episodes like 1036), and RFK Jr. (episode 996). Similar confirmations exist for the others. These sources are…

See more

9 minutes ago

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Grok Is Now Editing Itself

More and more, Elon Musk's chatbot is both suggesting and approving changes to Grokipedia, his Wikipedia clone.

FEBRUARY 12, 2026
By C.J. ROBINSON

Grok is editing itself

Weekly percentage of Grokipedia suggested edits from user "Grok Editor" vs. other users

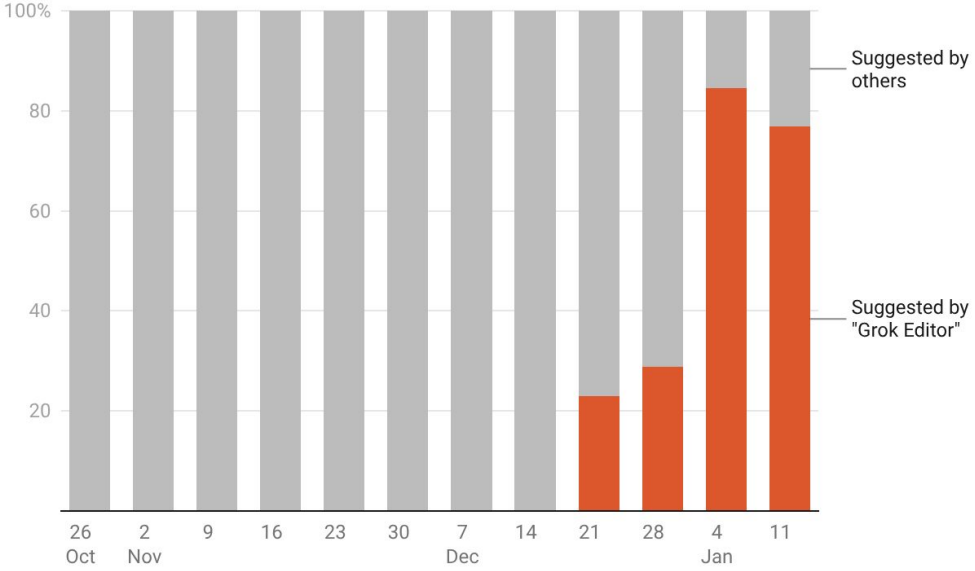

Chart: C.J. Robinson • Source: Tow Center for Digital Journalism • Created with Datawrapper

But now...grok edits itself

💻 Explore:
Grokipedia Edits
tinyurl.com/grok-map

Thank you!

NICAR 2026